

# Pricing the DeFi Tail: Do Protocols or Depositors Price Operational Risk?

**CBT 2026 — 10th International Workshop on**

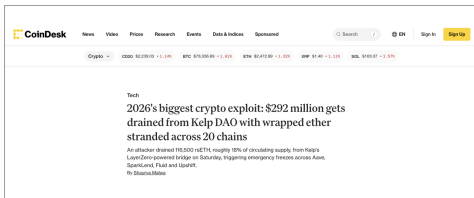
Cryptocurrencies and Blockchain Technology

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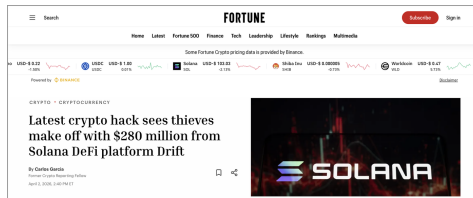
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The question

# Recent losses in DeFi



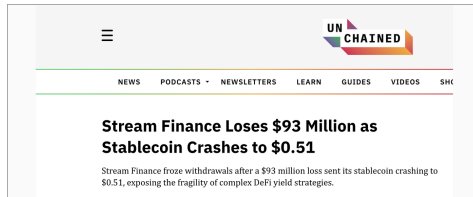
USD 292 m, Bridge — a forged cross-chain message



USD 290 m, Derivatives — admin approvals abused



USD 128 m, DEX — a rounding error, after audits



USD 93 m, Yield — no exploit at all

# DeFi removes many risks — operational risk is not one of them

DeFi's architecture genuinely removes several bank-balance-sheet risks: smart-contract custody replaces the custodian, settlement is atomic, counterparty credit risk collapses into overcollateralization.

- Operational risk remains — losses from failed processes, people, systems and external events — and in some cases concentrates: contract code, admin keys, oracles, economic design.
- Unlike a bank, DeFi is infrastructure: no regulatory capital requirement, no deposit insurance, no resolution authority. Nothing stands between a loss and the depositor's claim unless a protocol chooses to put it there.
- Yet DeFi markets earn products — lending supply, yield vaults, stablecoin positions, crypto neobanks, etc. — with minimal risk disclosures.

## Key takeaway

Not whether the risk exists — it does, and it is measurable — but *who bears it* and *whether anyone priced it*.

## Two responses, neither disciplined by a regulator

### **The protocol can hold a buffer.**

Aave's Umbrella module, the Sky surplus buffer and Compound's reserve factor are the DeFi's analogue to Basel capital reserves — just voluntary and sized by vote, not by any loss model. [BCBS 2006, 2017]

### **Or the depositor demands a premium.**

A supply yield above the risk-free rate compensates the depositor for bearing the tail — the market-discipline mechanism by which uninsured bank creditors price institutional risk. [Flannery 1998; Egan et al. 2017]

### Research question

Is either response sized to the operational-risk tail it is meant to cover? Answering it needs a common benchmark — so I first have to quantify the tail.

# Contributions

1. **A consolidated event dataset** — seven public feeds, deduplicated, filtered to depositor-facing DeFi losses, tagged by sector and Basel Level-1 event type.
2. **A per-sector Basel LDA**, fitted as a bank supervisor would, and benchmarked against banking operational risk.
3. **Both margins tested** against that one benchmark: protocol capital buffers and depositor risk premia.
4. **Policy discussion** — the standardized loss, yield and buffer disclosure a depositor needs in order to price the tail at all.

## Headline

Tails comparable to banking operational risk in four sectors and to cyber losses in three. Buffered venues cover  $\sim 5\%$  of their modeled  $\text{VaR}_{99.9}$  on average; unbuffered venues pay a higher yield than buffered ones, but it covers only  $\sim 1\%$  of their  $\text{VaR}_{99.9}$ .

Data

# Seven feeds, one deduplicated dataset

| Source                     | Raw   | In dataset |
|----------------------------|-------|------------|
| DefiLlama hack ledger      | 528   | 379        |
| kismp123 DSI corpus        | 823   | 290        |
| BlockSec Incidents Library | 285   | 201        |
| rekt.news leaderboard      | 281   | 219        |
| DeFiHackLabs catalog       | 615   | 163        |
| de.fi/rekt-database        | 4,030 | 704        |
| SlowMist Hacked            | 2,100 | 639        |
| Consolidated (dedup.)      |       | 1,164      |

- Median-across-sources loss reconciliation.
- Two-pass dedup: name within  $\pm 14$  d, then date  $\pm 7$  d and loss within  $\pm 10\%$ .
- Out-of-scope removed (CEX, memecoins, NFTs, phishing): 153 events.
- A depositor-facing filter drops 89 events (USD 0.84 B) whose loss the earn user does not bear.
- Working sample  $n = 1,075$ , USD 9.45 B; 50% cross-source confirmed.

## Why it matters

Prior work relies on a single feed, yet tail estimates are driven by the largest handful of events.

# Tagged to the banking taxonomy

Every event carries a DeFi sub-sector and a Basel Level-1 event type. Classified by textual evidence, with DefiLlama classification as fallback.

| Code | Basel Level-1                         | DeFi operational reading                                                      | Sector      |
|------|---------------------------------------|-------------------------------------------------------------------------------|-------------|
| IF   | Internal Fraud                        | Rugpull, owner-drain, insider exit-scam, upgradeable-contract abuse.          | Bridge      |
| EF   | External Fraud                        | Contract bugs (reentrancy, access control), key compromise, DNS hijack.       | Lending     |
| CPBP | Clients, Products, Business Practices | Economic-design attacks: flashloan governance, oracle and price manipulation. | DEX         |
| BDSF | Business Disruption, System Failures  | Chain-level: MEV, sequencer outage, consensus issue, chain halt.              | Yield       |
| EDPM | Execution, Delivery, Process Mgmt     | Team-side configuration and deployment errors.                                | Stablecoin  |
|      |                                       |                                                                               | Derivatives |
|      |                                       |                                                                               | Other       |

Employment Practices and Workplace Safety (EPWS) and Damage to Physical Assets (DPA) have no on-chain analogue and are omitted. [BCBS 2006]

# Sector and event type are orthogonal

Aggregate gross loss, USD m

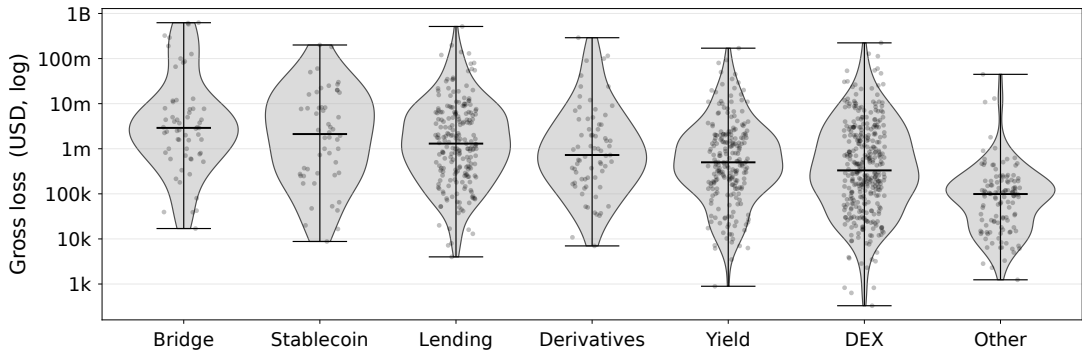
| Sector      | EF    | IF    | CPBP  | EDPM | BDSF | Total |
|-------------|-------|-------|-------|------|------|-------|
| Bridge      | 3,084 | 127   | 6     | 8    | 12   | 3,237 |
| Lending     | 704   | 568   | 928   | 14   | 8    | 2,222 |
| Stablecoin  | 178   | 105   | 456   | 27   | 8    | 773   |
| Yield       | 430   | 267   | 206   | 17   | 6    | 926   |
| DEX         | 1,044 | 136   | 221   | 0    | 1    | 1,402 |
| Derivatives | 148   | 0     | 275   | 2    | 380  | 805   |
| Other       | 35    | 4     | 48    | 0    | 0    | 87    |
| Total       | 5,623 | 1,207 | 2,140 | 68   | 415  | 9,452 |

- Aggregate mean loss USD 8.79 m, median USD 0.50 m, max USD 624 m (Ronin).
- Bridge and DEX are near-pure EF — code failing over pooled collateral. Bridge×EF alone is 33% of all losses.
- Lending and Stablecoin are CPBP-heavy — economic design, not code.

## Consequence for the model

Sector totals span two orders of magnitude and the loss mechanism differs by sector, so a single pooled fit would be meaningless. Everything is fitted per sector.

# Loss distributions differ sharply across sectors



Violins encode density, bars the median. Bridge carries the highest median event loss (USD 2.90 m) from the pooled collateral bridges hold.

# Method

# The loss-distribution approach, per sector

**Severity** — peaks-over-threshold GPD on excesses  $Y_i = X_i - u \mid X_i > u$ :

$$g_{\xi, \beta}(y) = \frac{1}{\beta} \left(1 + \xi \frac{y}{\beta}\right)^{-(1/\xi+1)}$$

The threshold  $u$  sits at the quantile  $q^*$  where  $\hat{\xi}$  plateaus, subject to  $n_u \geq 20$  and  $n_u/n \geq 10\%$ . The severity  $F_X$  is then piecewise: the empirical body below  $u$ , the GPD tail above it.

**Frequency** — monthly negative binomial  $N_s \sim \text{NB}(\mu_s, \alpha_s)$ ; annual counts as twelve iid monthly draws.

**Compound aggregation**

$$S_{\text{year}} = \sum_{i=1}^N X_i, \quad N \sim \text{NB}(\hat{\lambda}, \hat{\alpha}), \quad X_i \stackrel{\text{iid}}{\sim} F_X$$

$2 \times 10^5$  simulated years per sector, each event capped at the largest single-protocol exposure.

$\mathbb{E}[S]$  is the actuarial pure premium;  $\text{VaR}_{99.9}$  is the capital buffer the Basel AMA would compute. Both expressed as a share of trailing-365d TVL — the on-chain analogue of assets on books.

# From sector to protocol

No single protocol records enough events for its own fit — the same problem Basel solves for data-poor banks with a business-line severity allocated by an exposure indicator.

- Fit once per sector; treat its protocols as sharing that loss process per unit of TVL (credibility-theory shrinkage).
- A sector event strikes protocol  $p$  with probability  $\text{TVL}_p/\text{TVL}_s$ , so  $\lambda_p = (\text{TVL}_p/\text{TVL}_s) \lambda_s$ , same severity, capped at  $p$ 's own exposure.
- The tail does *not* thin linearly: with the single-loss approximation  $\text{VaR}_{99.9}(S_p) \approx F_X^{-1}(1 - 0.001/\lambda_p)$  scaling as  $\text{TVL}_p^{\hat{\xi}}$ ,  $\hat{\xi} < 1$ , the VaR-to-TVL ratio rises as a venue shrinks.

## Key takeaway

Each venue inherits its sector's loss process in proportion to its TVL, which is what makes a per-protocol figure possible without a per-protocol history. And because the tail does not thin linearly, that figure has to be simulated venue by venue rather than scaled from the sector ratio.

The tail

## Severity: four sectors like a bank, three like cyber

| Sector      | $n$ | $q^*$ | $n_u$ | $\hat{\xi}$ [95% CI] | $\hat{\beta}$ |
|-------------|-----|-------|-------|----------------------|---------------|
| Bridge      | 61  | 0.65  | 20    | +1.87 [+0.58, +3.18] | 17.6          |
| Lending     | 207 | 0.90  | 21    | +0.75 [-0.28, +1.50] | 24.8          |
| Stablecoin  | 52  | 0.60  | 21    | +0.65 [-0.39, +1.33] | 13.2          |
| Derivatives | 71  | 0.65  | 25    | +1.45 [+0.42, +2.38] | 3.7           |
| Yield       | 222 | 0.90  | 23    | +0.61 [-0.22, +1.24] | 11.5          |
| DEX         | 341 | 0.85  | 51    | +0.72 [+0.18, +1.15] | 7.1           |
| Other       | 121 | 0.80  | 24    | +1.58 [+0.54, +2.63] | 0.2           |

- Lending, Stablecoin, Yield, DEX cluster in  $\hat{\xi} \in [0.61, 0.75]$  — no heavier than the Moscadelli banking band  $[0.85, 1.39]$ , and all finite-mean. [Moscadelli 2004]
- Bridge, Derivatives, Other reach  $\hat{\xi} \approx 1.5\text{--}1.9$  — cyber-loss level, past the infinite-mean boundary. [Eling & Wirfs 2019]
- $\hat{\xi}$  sets how heavy the tail is;  $\hat{\beta}$  is the GPD scale in USD m. A Vuong test cannot separate GPD from lognormal in any sector, so I keep the GPD. [Vuong 1989]

## Frequency: up to five events a month

| Sector      | $n$ | $\hat{\mu}_s$ | $\hat{\lambda}_s$ | $\hat{\alpha}_s$ | $p$                   |
|-------------|-----|---------------|-------------------|------------------|-----------------------|
| Bridge      | 61  | 0.80          | 9.7               | 0.72             | 0.013                 |
| Lending     | 207 | 2.72          | 32.9              | 0.26             | 0.0014                |
| Stablecoin  | 52  | 0.68          | 8.3               | 0.14             | 0.590                 |
| Derivatives | 71  | 0.93          | 11.3              | 0.41             | 0.054                 |
| Yield       | 222 | 2.92          | 35.3              | 0.60             | $7.7 \times 10^{-11}$ |
| DEX         | 341 | 4.49          | 54.2              | 0.31             | $5.4 \times 10^{-10}$ |
| Other       | 121 | 1.59          | 19.2              | 0.89             | $3.6 \times 10^{-8}$  |

- Annual rates  $\hat{\lambda}_s$  span sixfold: 8.3/yr (Stablecoin) to 54.2/yr (DEX), with Lending at 32.9/yr.
- Dispersion  $\hat{\alpha}_s$  is positive everywhere, 0.14 (Stablecoin) to 0.89 (Other): events cluster rather than arrive independently. Cross-sector index  $D = 3.27$ .
- A likelihood-ratio test rejects Poisson at  $p \leq 0.02$  in five of seven sectors — under it the annual aggregate, and its  $\text{VaR}_{99.9}$ , would be understated.

# The capital the tail implies

VaR<sub>99.9</sub>, % of trailing-365d TVL

| Sector      | <i>n</i> | TVL (B) | VaR <sub>99.9</sub> | VaR <sub>99.9</sub> <sup>U</sup> | IQR          |
|-------------|----------|---------|---------------------|----------------------------------|--------------|
| Bridge      | 61       | 49.1    | 32.5                | 67,326                           | [25.5, 40.8] |
| Lending     | 207      | 66.8    | 18.0                | 22.3                             | [3.2, 18.9]  |
| Stablecoin  | 52       | 26.5    | 16.1                | 16.1                             | [3.1, 30.1]  |
| Derivatives | 71       | 65.9    | 22.8                | 638.0                            | [21.8, 27.1] |
| Yield       | 222      | 11.4    | 15.3                | 29.5                             | [5.5, 17.3]  |
| DEX         | 341      | 14.5    | 14.1                | 46.2                             | [10.5, 15.9] |
| Other       | 121      | 18.2    | 27.5                | 285.9                            | [4.8, 27.7]  |

- Core sectors need 14–18% of TVL — against about 0.6% of assets for a bank, so 25–30× as much. [de Fontnouvelle et al. 2007]
- Bridge, Derivatives, Other look larger (23–32%) but only because each event is capped at exposure.
- Uncapped (VaR<sub>99.9</sub><sup>U</sup>),  $\hat{\xi} > 1$  has no stable quantile: Bridge diverges to 673× TVL.

## The familiar pathology

The heaviest-tailed, most consequential sectors are exactly the ones the model cannot pin down unaided — the instability that retired Basel's internal-model AMA in 2017.

# Sensitivity analysis

**Sensitivity to the largest events.** Dropping the single largest event moves  $\hat{\xi}$  by 0.16–0.5, most sharply in Lending (0.75  $\rightarrow$  0.35). Under any drop-top rule up to 0.5%, every sector keeps its below / above infinite-mean character.

**Sensitivity to the cap.** The cap binds: removing it lifts Lending 18  $\rightarrow$  22%, DEX 14  $\rightarrow$  46%, Yield 15  $\rightarrow$  30%. Only Stablecoin is cap-insensitive. The  $\hat{\xi} > 1$  sectors are stable only under the cap — uncapped, Bridge diverges to  $673\times$  TVL and Derivatives to 638%.

**Sensitivity to the confidence level.** At 99.5% and 99% the Lending requirement falls to 7% and 5% of TVL, and buffered-venue coverage rises from 5% to 18% and 31%.

## What survives, and what does not

The  $\text{VaR}_{99.9}$  level is genuinely sensitive — most of all to the cap. What does not move is the conclusion: every sector keeps its tail character, and removing the cap only raises the requirement, so both margins stay an order of magnitude short under all three tests.

Who bears the tail

# The protocol's response: capital buffers

Top-10 Lending venues by gross supplied TVL (DefiLlama, June 2026): four disclose an operational-risk reserve, six do not.

| Protocol        | TVL (B) | Buffer (B) | $\mathbb{E}[S]$ (B) | VaR <sub>99.9</sub> (B) |
|-----------------|---------|------------|---------------------|-------------------------|
| Aave V3         | 11.85   | 0.363      | 0.076 (475%)        | 4.00 (9%)               |
| SparkLend       | 3.41    | 0.083      | 0.021 (390%)        | 1.58 (5%)               |
| Compound V3     | 1.04    | 0.043      | 0.006 (700%)        | 0.64 (6%)               |
| Venus Core Pool | 0.98    | 0.011      | 0.005 (187%)        | 0.61 (2%)               |

Parentheses give the buffer's coverage of each. Average coverage of the modeled capital buffer:  $\sim 5\%$ , a 95% shortfall, from 2% (Venus) to 9% (Aave V3).

## Key takeaway

All four buffers cover the pure premium — but fall well short of the tail. They are sized to an average year, not to the 1-in-1,000 year banking regulation asks a bank to hold capital against. [BCBS 2006, 2017]

# The depositor's response: the yield premium

| Protocol           | Total yield | Risk premium | $E[S]$         | $VaR_{99.9}$    |
|--------------------|-------------|--------------|----------------|-----------------|
| <i>No buffer</i>   |             |              |                |                 |
| Morpho Blue        | 4.60 %      | +90 bps      | 62 bps (144%)  | 3,905 bps (+2%) |
| JustLend V1        | 3.42 %      | -28 bps      | 60 bps (-47%)  | 4,785 bps (-1%) |
| Kamino Lend        | 2.91 %      | -79 bps      | 55 bps (-143%) | 6,169 bps (-1%) |
| Jupiter Lend       | 4.64 %      | +94 bps      | 55 bps (172%)  | 6,605 bps (+1%) |
| Fluid Lending      | 6.27 %      | +257 bps     | 52 bps (497%)  | 6,507 bps (+4%) |
| Euler V2           | 3.59 %      | -11 bps      | 47 bps (-23%)  | 8,131 bps (0%)  |
| <i>With buffer</i> |             |              |                |                 |
| Aave V3            | 3.36 %      | -34 bps      | 64 bps (-53%)  | 3,378 bps (-1%) |
| SparkLend          | 2.84 %      | -86 bps      | 60 bps (-143%) | 4,634 bps (-2%) |
| Compound V3        | 2.85 %      | -85 bps      | 55 bps (-155%) | 6,101 bps (-1%) |
| Venus Core Pool    | 2.09 %      | -161 bps     | 55 bps (-295%) | 6,226 bps (-3%) |

Risk premium = trailing-30d TVL-weighted supply APY minus the 3.70% T-bill. Unbuffered venues average +54 bps — about their 55-bps pure premium, but only 1–4% of the tail. Protocol and depositor alike size to an average year, not the extreme one.

## Both margins together

Because buffer and premium substitute, an efficient market would trade one for the other — buffered venues paying less. The data run in that direction.

- Median premium:  $-86$  bps with a buffer against  $+39$  bps without — a 125-bps gap. One-sided Mann–Whitney  $p = 0.01$ , and  $p = 0.03$  extending to all Lending venues with net TVL  $\geq$  USD 100 m ( $n = 19$ ).
- Below that size the signal attenuates: posted yields are dominated by airdrops and incentives. The long-tail depositor is chasing incentives, not pricing risk.

### Right direction, wrong magnitude

The whole premium cross-section spans 418 bps, while the tail asks for about 18% of TVL. The market prices operational risk — and still leaves the tail overwhelmingly unfunded on both margins.

What follows

# Policy

## The shortfall is not evenly borne.

- A professional participant can assess the tail, negotiate bilateral terms through governance, and size or exit the position.
- A retail depositor sees only the posted rate and bears the unpriced tail disproportionately. [Cornelli et al. 2025]
- Market discipline presumes the depositor *can* price the risk. For retail that presumption fails twice: the information is not disclosed, and pricing a heavy tail takes skills they do not have. [Flannery 1998]

## Disclosure levels the playing field.

- Capital mandates have no clear path onto autonomous protocols — and supervisors themselves retired the internal-model approach. [BCBS 2017]
- Standardized loss-and-yield disclosure — a Pillar 3 analogue, per sector, alongside the posted yield.
- Buffer-and-tail disclosure — the buffer, the share of  $\text{VaR}_{99.9}$  it covers, the residual the depositor bears.
- The duty attaches to whoever fronts access: front-ends, wallets, aggregators, custodial products.

# Conclusion

## What I find.

- A measurable tail: USD 9.45 B over 1,075 events; four core sectors no heavier-tailed than banking, three past the infinite-mean boundary.
- Calibrated as a supervisor would, the Lending tail asks for  $\sim 18\%$  of TVL.
- Against it, buffers cover 5% and unbuffered venues pay +54 bps — short by an order of magnitude.
- The two margins move together, so the market discriminates; it just does not price.

## Limitations.

- **Data.** Publicly reported losses only, and gross of recovery — so the estimates are conservative.
- **Homogeneity.** One severity per sector, allocated by TVL share.
- **Identification.** Buffered venues are also the largest and oldest, and the yield spread is reduced-form — an association, not an identified effect.
- **Stationarity.** 2020–2026 fitted as one regime.

## The recommendation

Not capital mandates, but disclosure: standardized loss, yield and buffer-coverage information, owed by whoever fronts access.

# Questions and discussion.

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